

be considered, which is approximately two years. Therefore, before using any capacitor, carefully check the capacitance value and its remaining life.

The reason why capacitors should be kept cool is that most SMPS inside cabinets are exposed to 45-50 degrees Celsius temperature. At that point, the failure rate of a capacitor increases drastically. By keeping the ambient temperature inside the cabinet/capacitor lower, the chances of failure decrease. Also, if you are using a capacitor at a higher temperature, then make sure that the capacitor is not very old.

When it comes to capacitor longevity, we generally go by whatever temperature is mentioned in the datasheet. However, we rarely check the reason behind its specification.

As per Arrhenius' Law, the life of a capacitor becomes half with every 10°C increase in the ambient temperature. So, it should be a common practice to make the lifecycle calculation after the prototype by measuring the ambient temperature (and not just go

by the specification mentioned in the datasheet). Only after that the SMPS lifecycle should be finalised.

Overlooked reasons for MOSFET failure

Some common reasons for MOSFET losses are:

- High drain current and voltage
- Incorrect heat-sink placement
- Shorting of legs

Problems can occur in MOSFETs due to a high rate of rise in drain-to-source voltage, causing a MOSFET to switch ON while it is switching OFF, which is dangerous. This happens because the gate threshold voltage typically reduces to less than 70% of its 25°C value at the maximum junction temperature. So, while designing, the important parameter to consider is temperature. The gate threshold voltage should be taken at the working temperature of the MOSFET and accordingly the ON-OFF switching process designed, so that no false triggering of the MOSFET occurs (while it is turning OFF). This will reduce the MOSFET failure to a great extent.

Other factors to keep in mind during placement of MOSFET in the design are:

- PCB creepage factor
- Correct placement of and height of legs near heatsink and in PCB
- Correct package
- Correct snubber design to avoid peak voltages

Lossless snubber design

As seen from the graph in Fig. 1, without snubber the spike affects the

circuit and damages the power switch. To mitigate this effect, a snubber circuit is used.

Snubbers are devices for suppressing voltage spikes and thus controlling the rate of voltage change. Clamps, on the other hand, are used only for voltage spikes and do not help in EMI. So, we choose them accordingly. They both have their uses, trade-offs, and circuit requirements.

You must have used snubbers several times. But I doubt you ever calculated the losses in them. Generally, we engineers tend to think that snubbers are here to help us. They are often thought to be more as saviours than culprits of power losses, but, in reality, the losses caused by them silently kill the circuit efficiency. So, if you have never calculated power losses that occur due to snubbers, don't worry. Let's check them here.

The formula for calculating power loss due to snubbers is:

$$Pr_s = 0.5 \cdot V_s^2 \cdot C_s \cdot f_{sw}$$

where V_s = Voltage across the snubber capacitor C_s

f_{sw} = Switching frequency

There are two ways to achieve more efficiency in already-optimised circuits: either optimise the snubbers or use a lossless snubber.

A lossless snubber is a snubber whose energy is trapped and recovered by the power circuits. It is designed to absorb a fixed amount of energy and store that in a capacitor.

As seen in Fig. 2(a), the size of the capacitor decides how much energy the snubber can absorb and then give the same back to the circuit. The clamp circuit works by first absorbing the leakage energy in the clamp capacitor and then recycling it through an additional transformer winding, called the snubber winding. So, theoretically, no losses occur and whatever energy that has been captured is given back to the main circuit.

At the primary MOSFET turn OFF, the clamp capacitor is recharged until V_{clamp} reaches the secondary voltage reflected to the primary side.

In diagram (b), it can be seen that if V_{clamp} becomes too high, the excess energy is recycled to the output rail and the input bus (shown by the solid and dotted red-colour arrows).

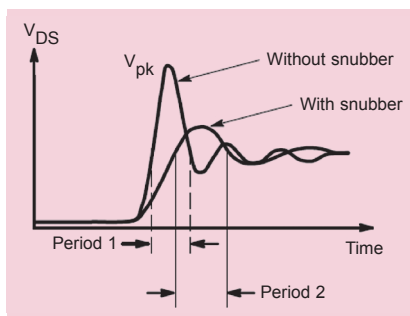


Fig. 1: Effects with and without snubber (Source: AN1695/D)

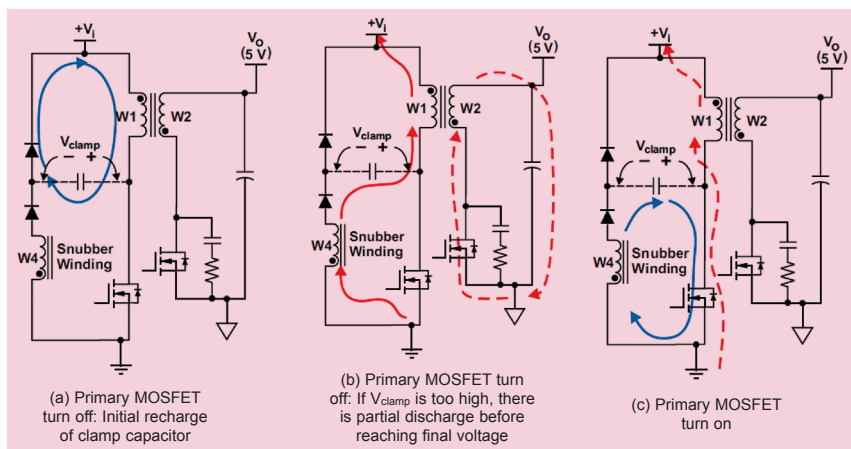


Fig. 2: Snubber circuit (Source: Under the hood of Flyback SMPS by TI)